

Fleet Management System

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Commercialization of Mobile Robots

Robot Vacuum

- Comm.: Mid-2000s
- Adoption: ~2010+



Serving Robots

- Comm.: Early 2010s
- Adoption: ~2020+



Logistics Robots

- Comm.: Early 2010s
- Adoption: ~2020+



Manufacturing Robots (Mobile Manipulators)

- Comm.: ~2025
- Adoption: ≤2030



Service Robot (Humanoid)

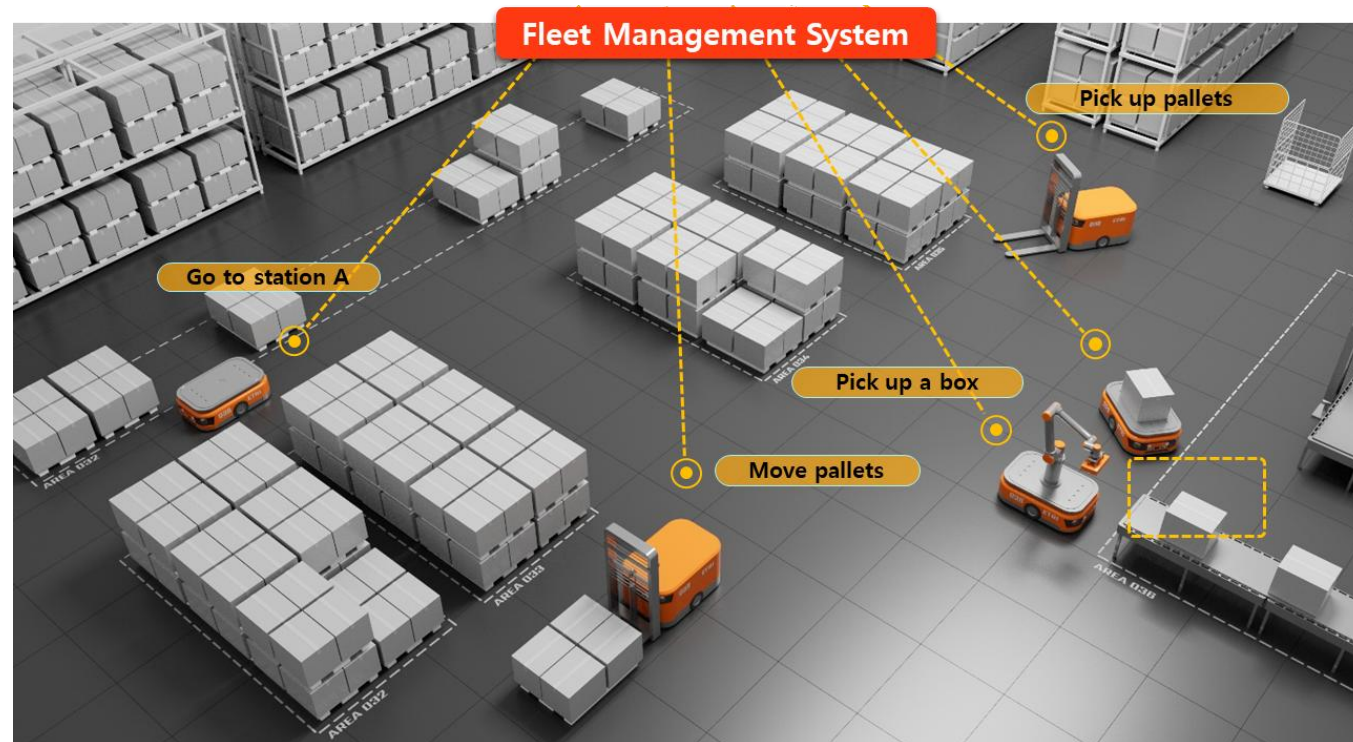
- Comm.: ≥2030



Introduction

로봇 편대관리시스템 (FMS: Fleet Management System)

- 서버에 구축이 되어 물리세계의 로봇에게 작업을 할당하고 작업을 모니터링하며, 물동량 흐름을 제어하는 역할



Introduction

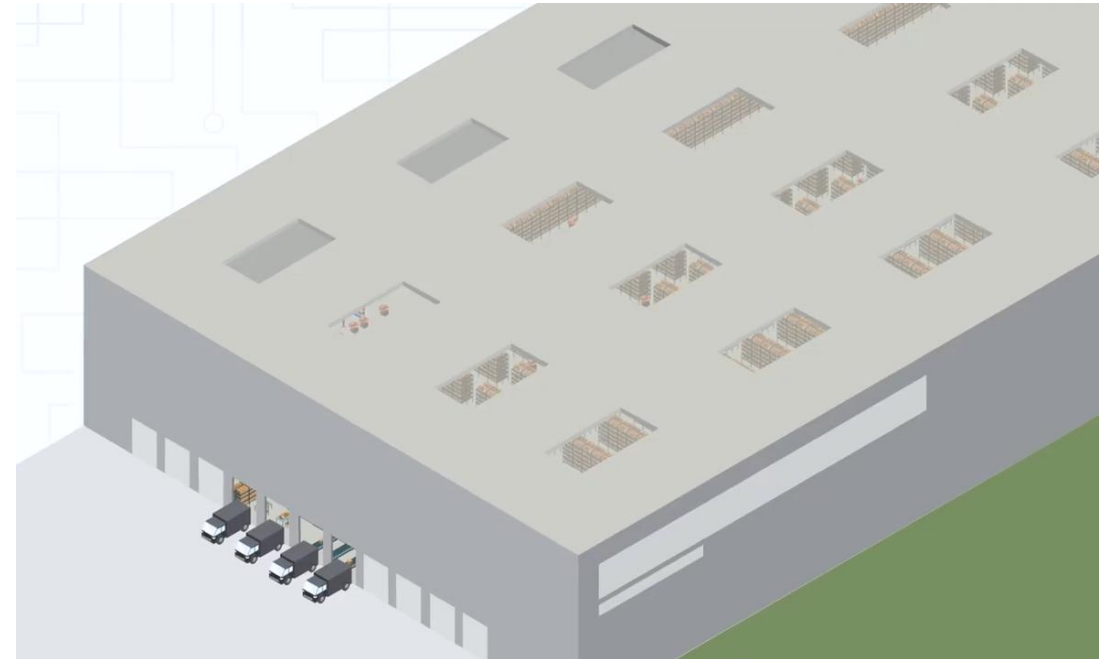
■ FMS 최근 사례 및 핵심 기술이슈

- 확장성: **다양한 로봇** (AGV, AMR, AGF), **다양한 공간** (A사, B사, C사), **다양한 물류 작업 시나리오** (KIVA 시스템, Order Picking, Batch Picking, Zone Picking, Stacking)에 빠르고 효과적으로 대응

아마존 KIVA 로봇 시스템 [1]



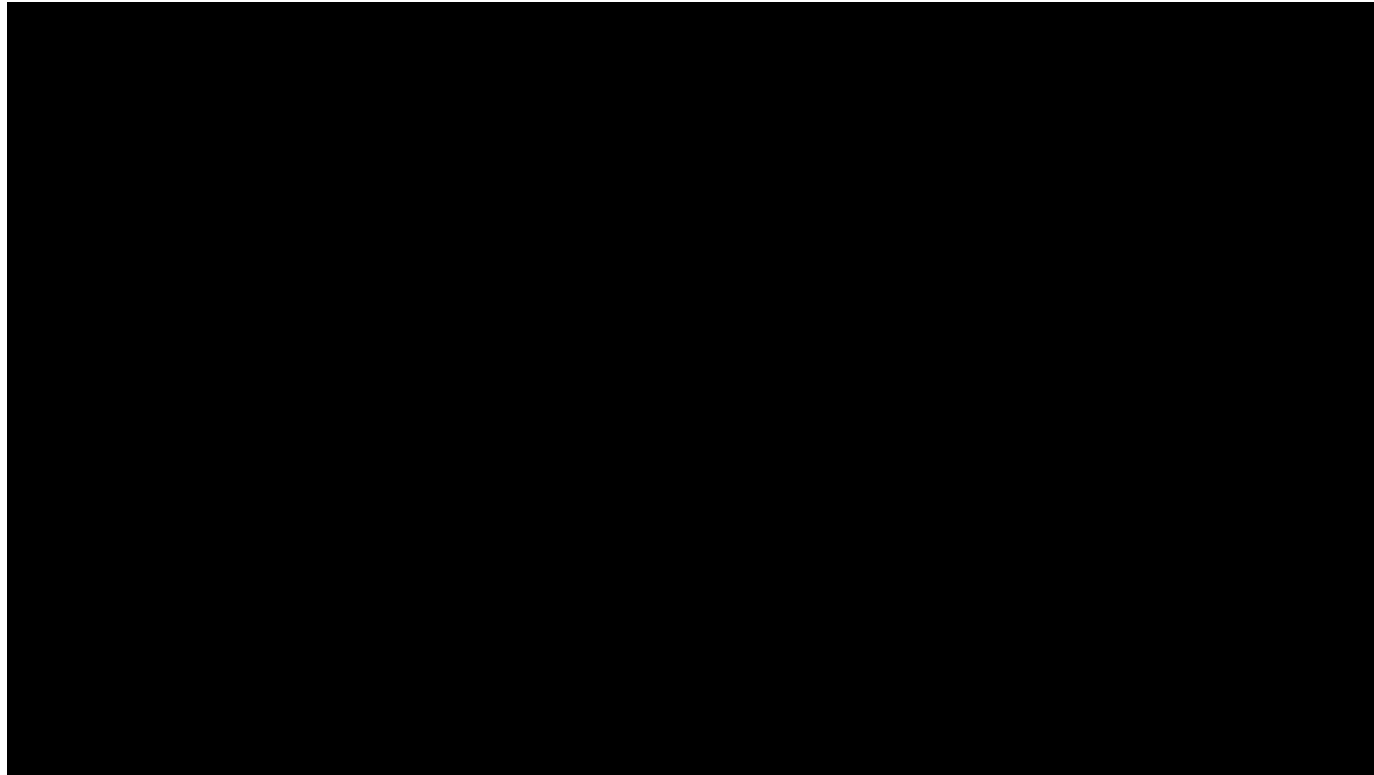
inVia 로보틱스 Order Picking [2]



Introduction

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Introduction

■ FMS 최근 사례 및 핵심 기술이슈

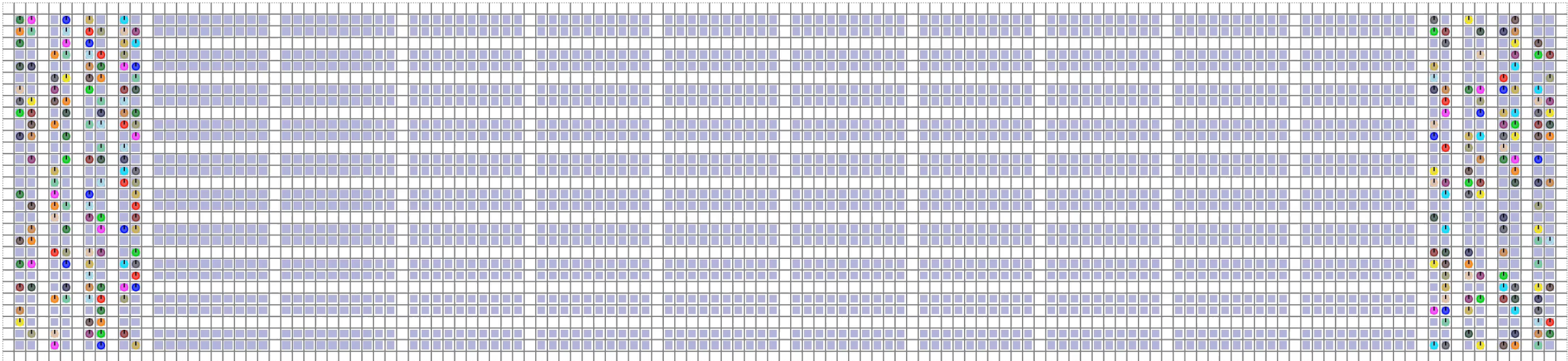
- 기술력 확보 필요: 우리나라는 물류 로봇 관련 핵심 하드웨어와 기술 대부분 해외에 의존



Issues

Problems

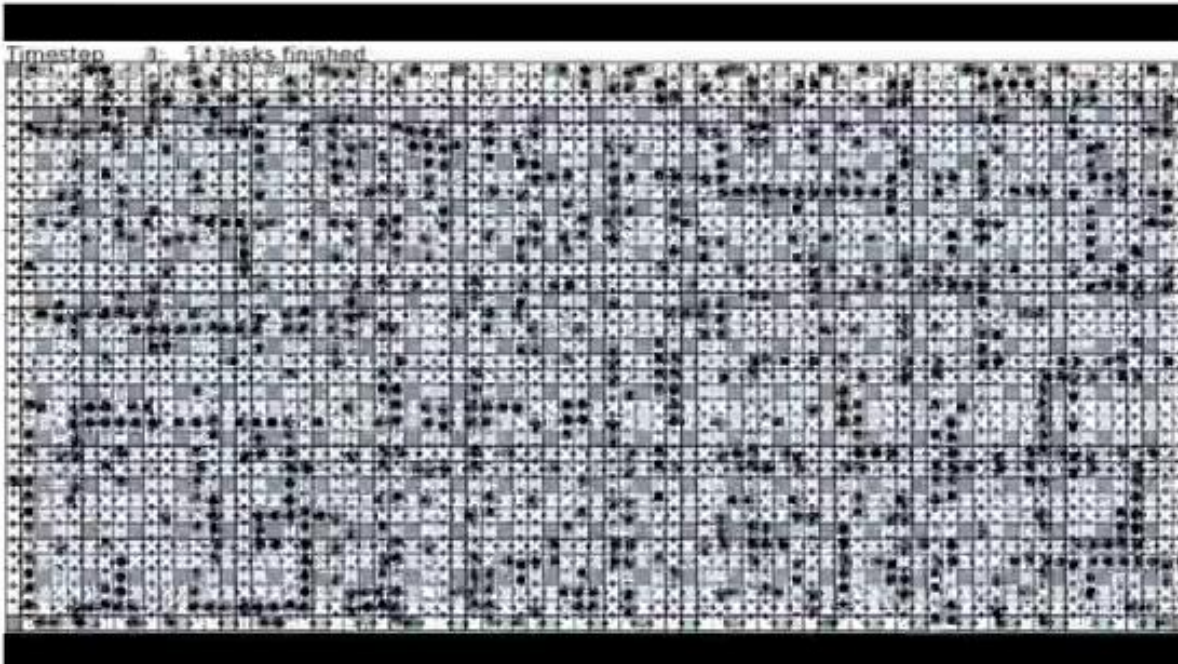
- **Multi-agent task assignment:** tasks should be assigned to robots in an optimal sequence to maximize the throughput
- **Multi-agent path finding:** robots must arrive their goal locations quickly without colliding with each other



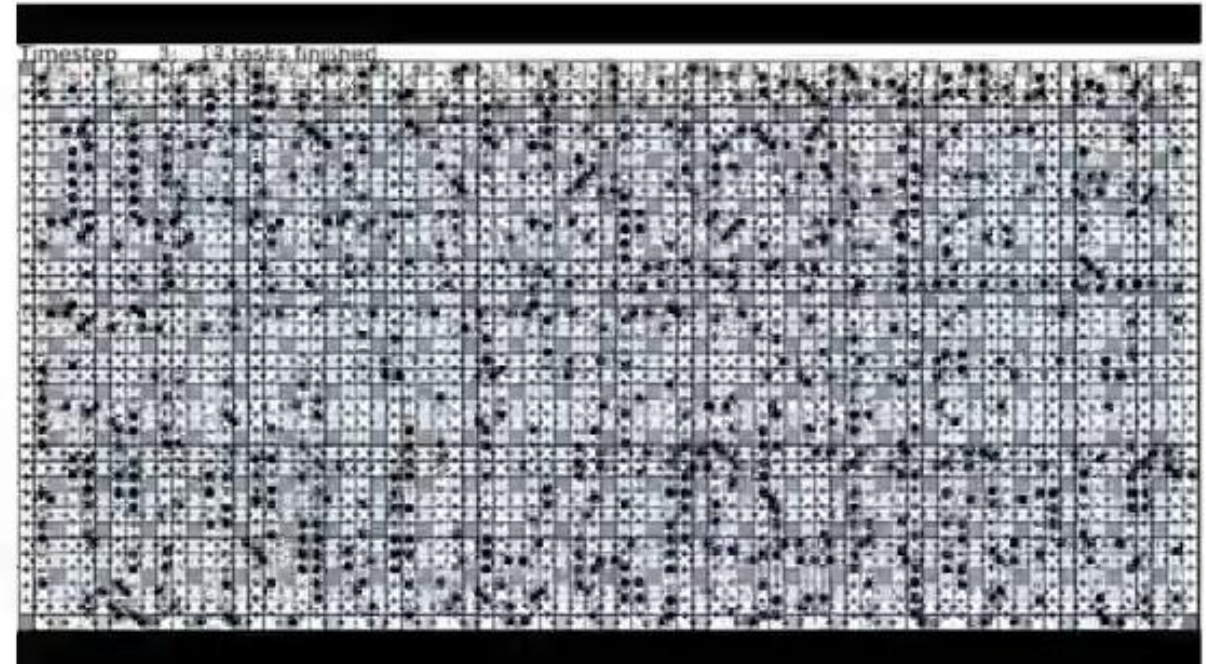
Issues

Multi-agent path finding

Single-Agent Planner



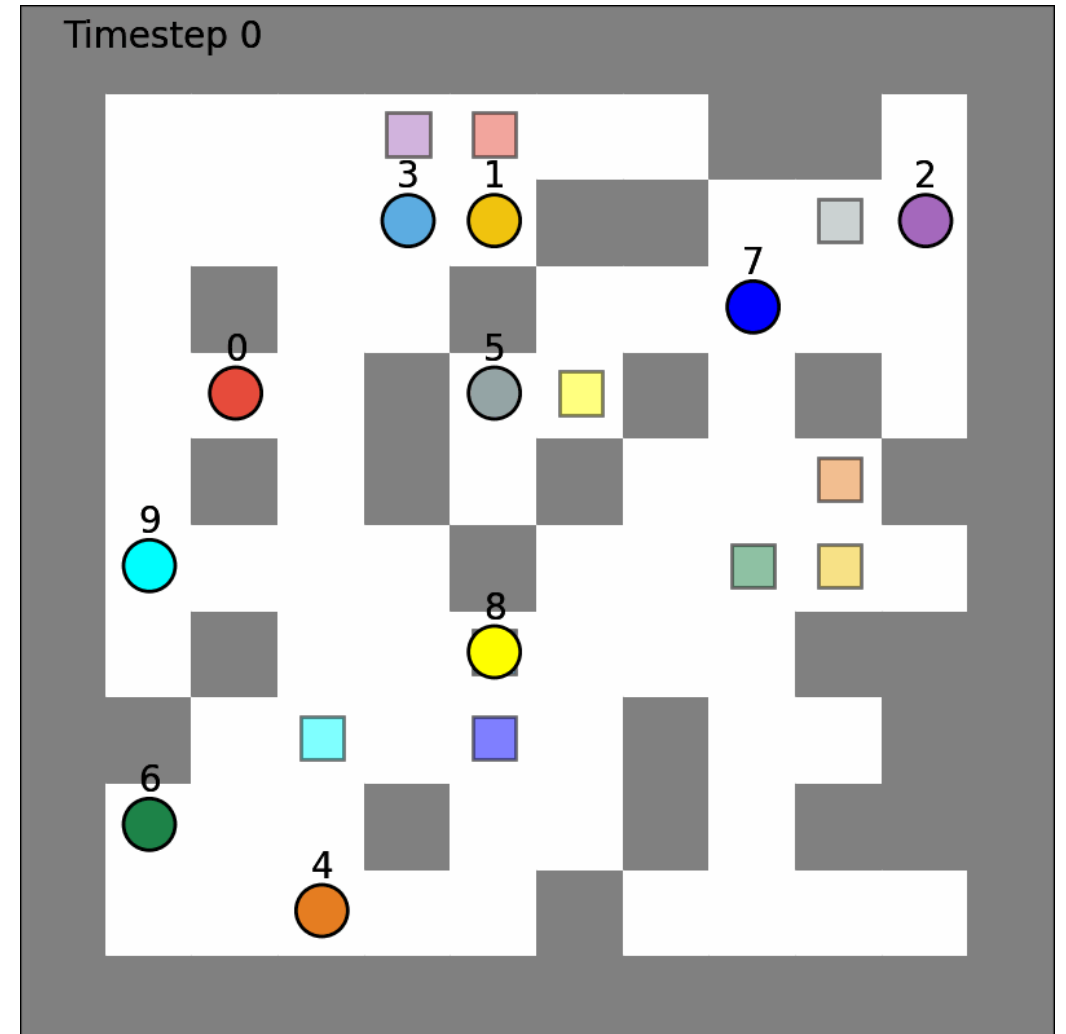
MAPF Planner



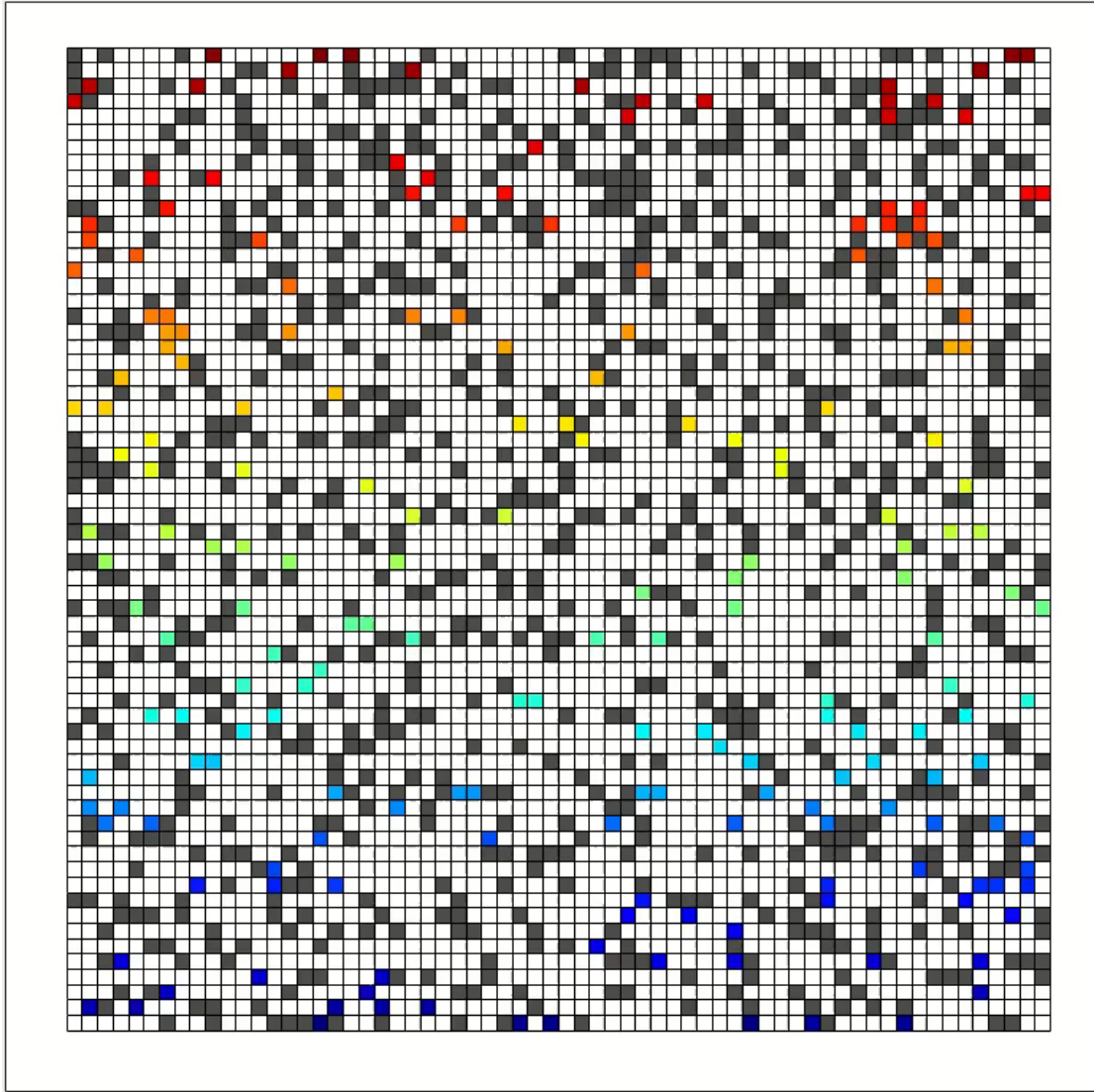
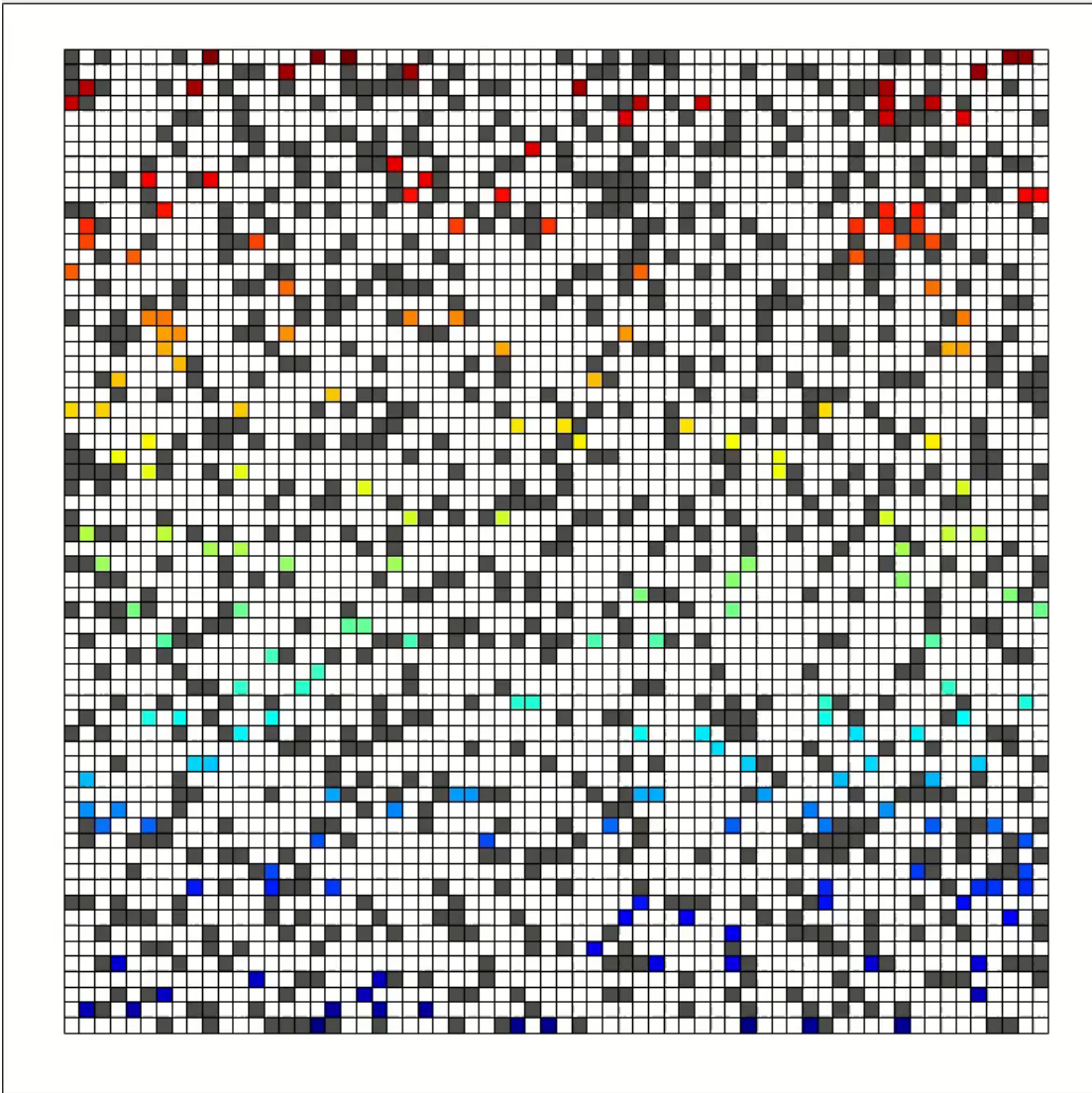
Multi-Agent Path Finding

Multi-agent path finding

- Given:
 - A grid map
 - A set of agents
 - Start and goal locations
- Goal:
 - Find collision-free paths
 - Minimize the total path costs
- Assume:
 - Move 4-directions or wait
 - No kinematic constraints



Multi-Agent Path Finding



Fleet Management System

Fleet Management System

FilePanelsHelp

TaskState

Task ID	Type	Start Node	Goal Node	State	Priority	Robot ID	Start Time
IN0	INBOUND	I440	R221	QUEUED	3		09:00:00
OUT1	OUTBOU...	R415	O443	QUEUED	3		09:00:00
IN2	INBOUND	I441	R129	QUEUED	3		09:00:00
OUT3	OUTBOU...	R223	O444	QUEUED	2		09:00:00
IN4	INBOUND	I442	R410	ACTIVE	1	AGF1	21:14:34
OUT5	OUTBOU...	R367	O445	ACTIVE	1	AGF3	21:14:36
IN6	INBOUND	I440	R119	ACTIVE	1	AGF0	21:14:33

☐ Active only

Refresh

Task ID

Type

Start Node

Goal Node

Priority

Reset

Submit

Response

☒ Auto Task ID

RobotState

Robot ID	Type	Mode	Task ID	Progress	Battery	Goal Node
AGF0	AGF	TO_START	IN6	31	100	R119
AGF1	AGF	TO_START	IN4	13	100	R410
AGF2	AGF	TO_START	OUT7	9	100	O443
AGF3	AGF	TO_START	OUT5	5	100	O445

Robot ID

Action

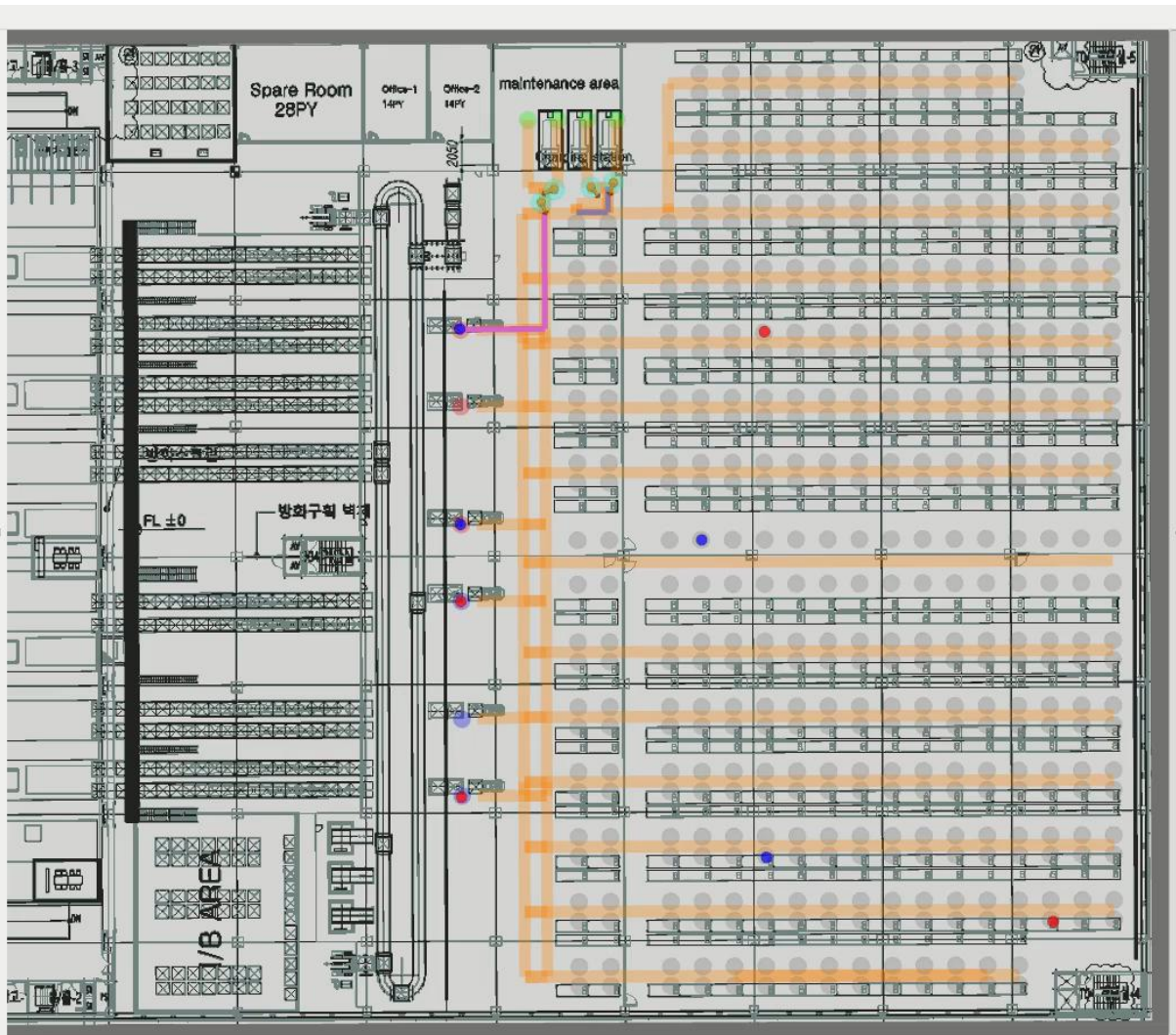
Goal Node

Reset

Response

Send

Reset



개별연구 1

현장실습-모바일 로봇 인공지능 기술 개발

- LLM(Large Language Model) 기술을 활용 다중 로봇 작업할당
- SMART-LLM: <https://sites.google.com/view/smart-llm/>

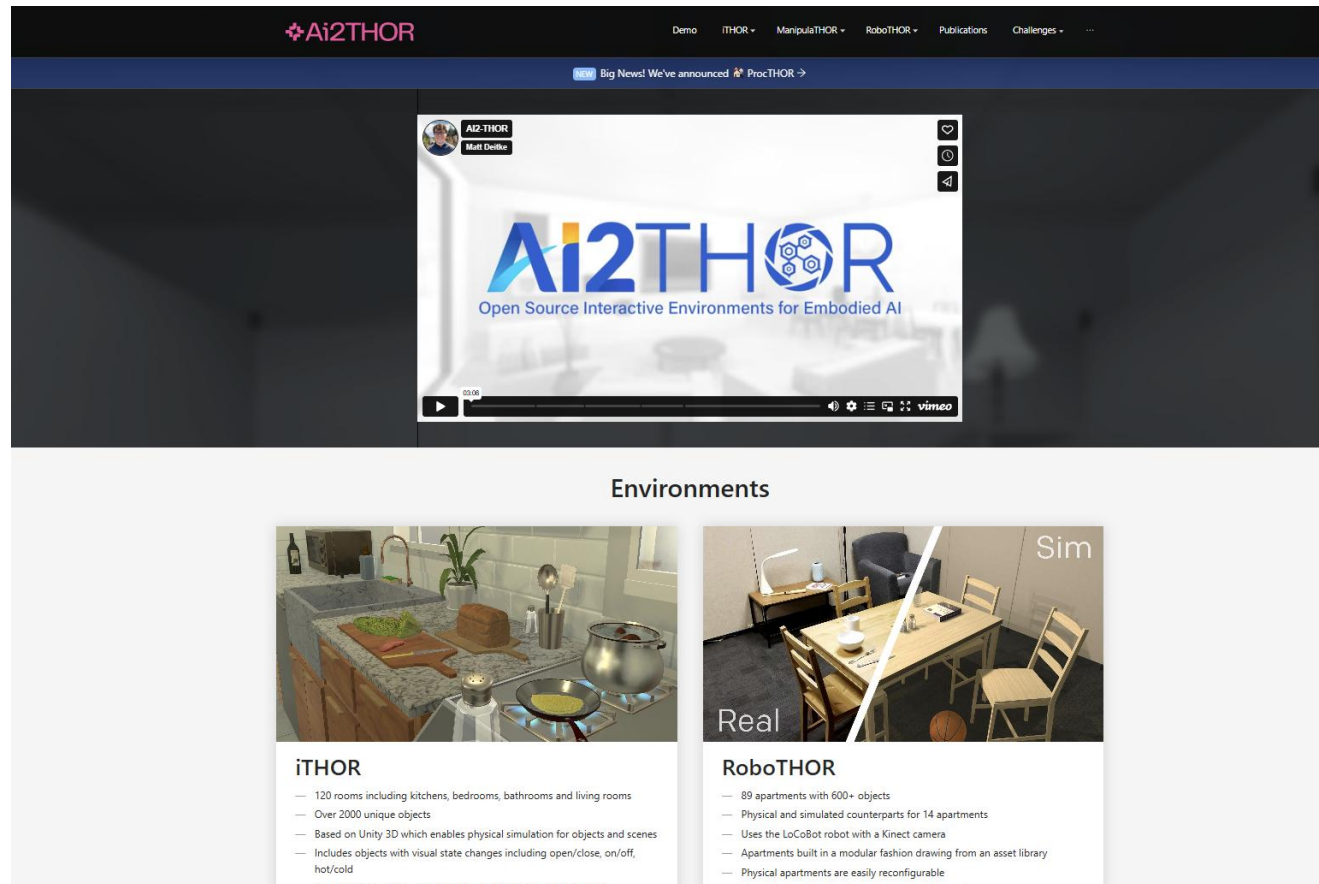


In this work, we introduce SMART-LLM, an innovative framework designed for embodied multi-robot task planning. SMART-LLM: Smart Multi-Agent Robot Task Planning using Large Language Models (LLMs), harnesses the power of LLMs to convert high-level task instructions provided as input into a multi-robot task plan. It accomplishes this by executing a series of stages, including task decomposition, coalition formation, and task allocation, all guided by programmatic LLM prompts within the few-shot prompting paradigm. We create a benchmark dataset designed for validating the multi-robot task planning problem, encompassing four distinct categories of high-level instructions that vary in task complexity. Our

개별연구 1

■ 현장실습-모바일 로봇 인공지능 기술 개발

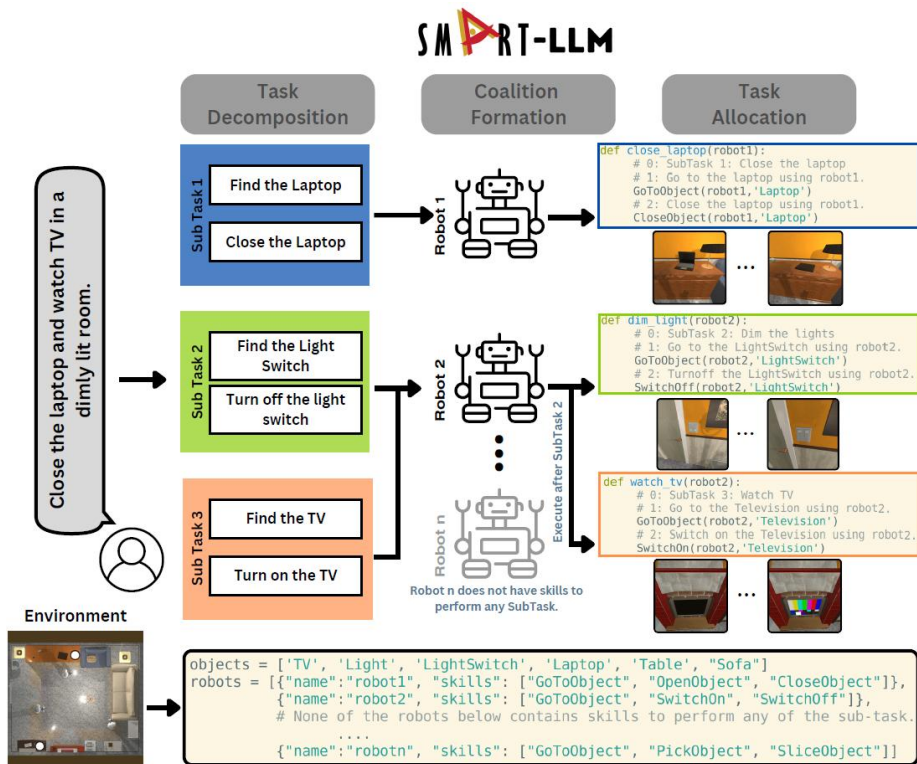
- iTHOR(AI2-THOR) 시뮬레이터 활용 다중 로봇 작업 수행 테스트 (<https://ai2thor.allenai.org>)



개별연구 1

■ 현장실습-모바일 로봇 인공지능 기술 개발

- SMART-LLM 기반 다중로봇 작업 할당 기술 개발
- LLM을 활용해 자연어 지시를 작업 분해 → 연합 구성 → 작업 할당 → 실행으로 변환
- 병렬 수행을 포함한 실행 가능한 Pythonic 코드를 자동 생성



[예시]

자연어명령: Break a Vase and turn on TV

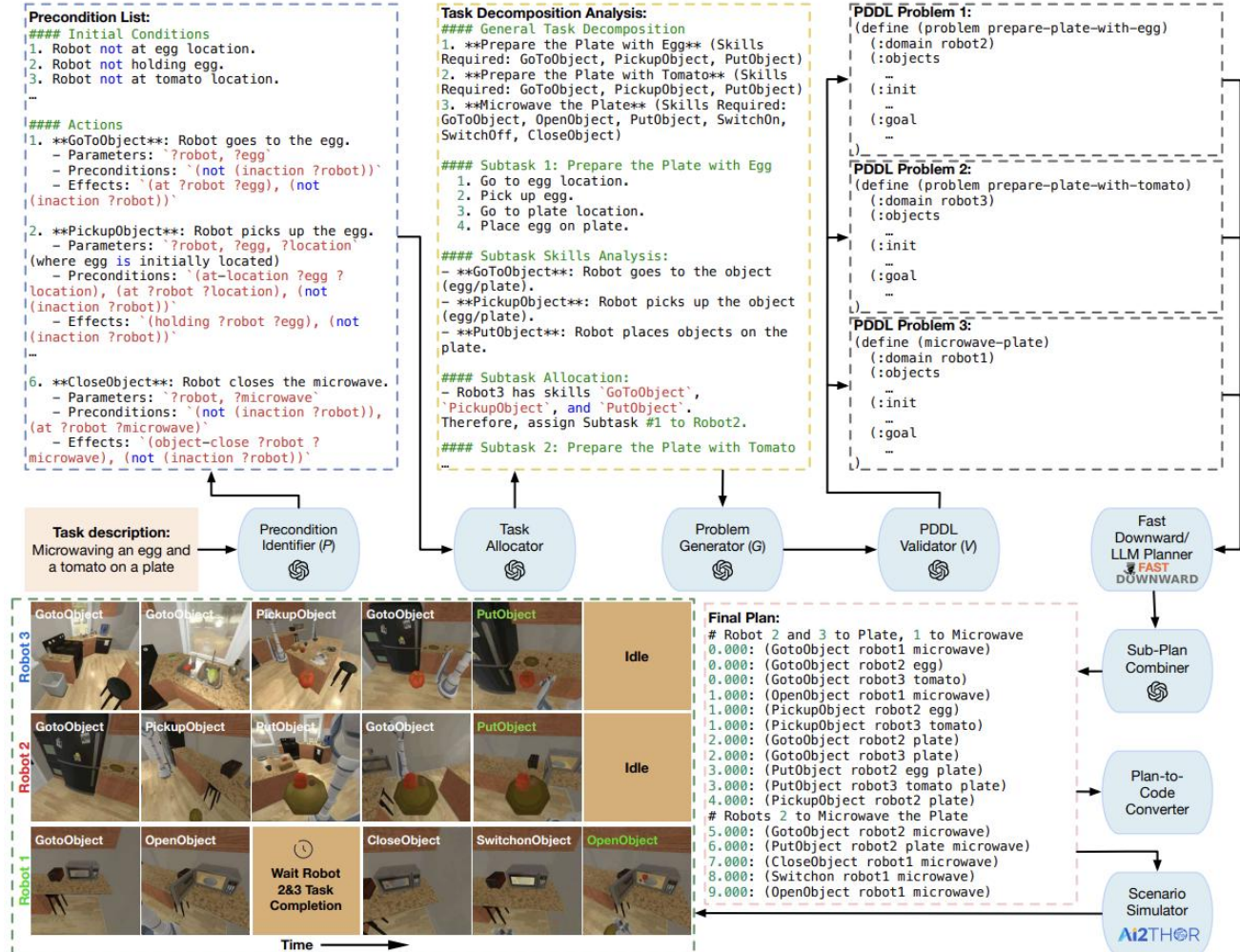
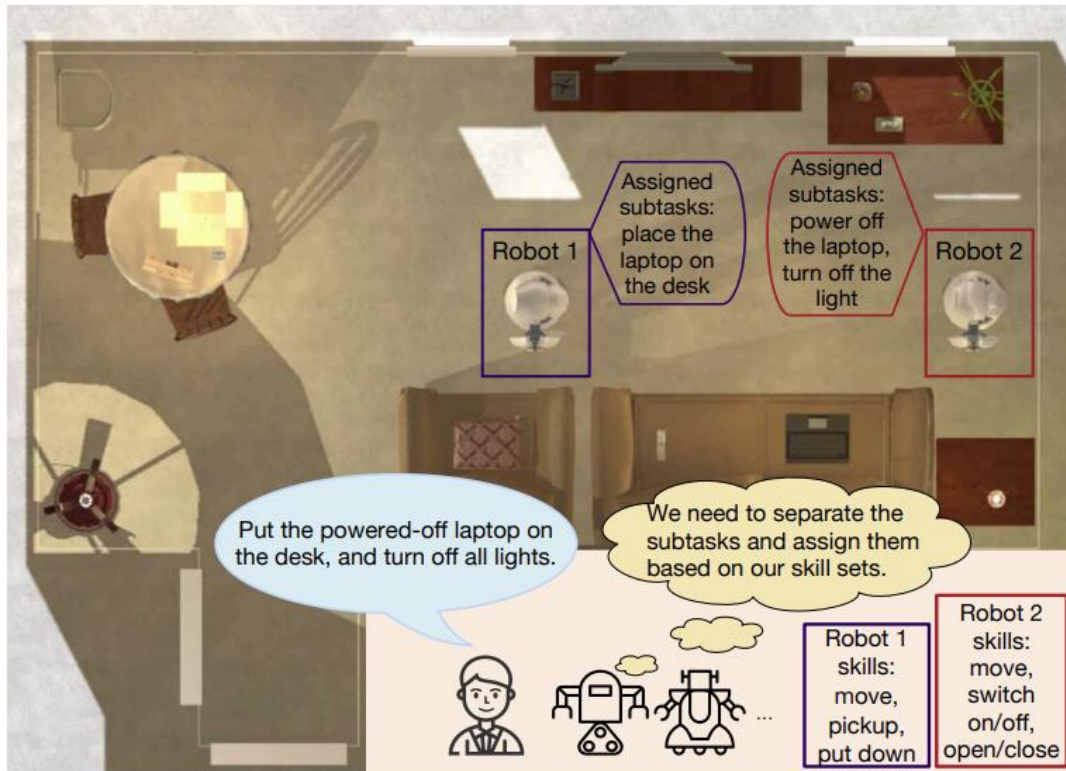
- 로봇1: Break a Vase
- 로봇2: Turn on TV



개별연구 1

현장실습-모바일 로봇 인공지능 기술 개발

- LaMMA-P (<https://lamma-p.github.io/>)



참고 논문

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- HiP-LR: Hierarchical Planning with Localized Failure Recovery for Long-Horizon Multi-Robot Tasks, Findings on EMNLP 2026

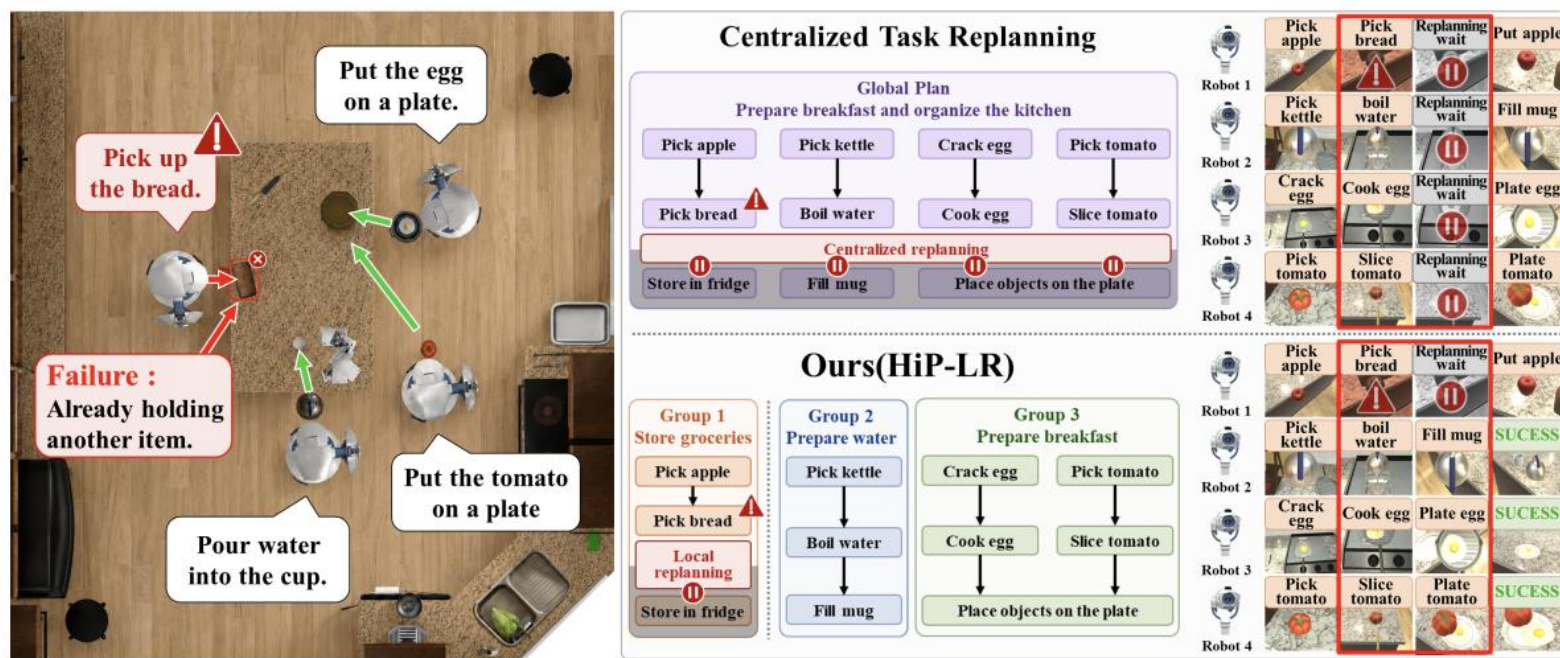


Figure 1: Example of failure-localized replanning in HiP-LR. Upon failure (e.g., occupied gripper), centralized methods (Shek et al., 2026) regenerate the global plan, halting all robots. Conversely, HiP-LR replans exclusively within the affected dependency-isolated group, allowing unaffected robots to continue executing in parallel.

MAP-THOR					
Method	SR↑	TR↑	C↑	B↑	SE↓
SMART-LLM	0.04	0.12	0.25	0.09	1.00
DMRS-1D	0.23	0.53	0.76	0.61	0.54
DMRS-2D	0.27	0.53	0.76	0.67	0.52
LaMMA-P	0.30	0.50	0.54	0.34	0.57
LaMMA-P+ [†]	0.46	0.65	0.69	0.33	0.60
H-AIM [†]	0.53	0.66	0.66	0.57	0.75
KGLAMP [†]	0.51	0.74	0.76	0.64	0.71
HiP-LR (Ours)[†]	0.73	0.90	0.87	0.75	0.48

RECAP-THOR					
Method	SR↑	TR↑	C↑	B↑	SE↓
LaMMA-P+ [†]	0.25	0.50	0.56	0.21	0.72
KGLAMP [†]	0.12	0.60	0.65	0.10	0.66
HiP-LR (Ours)[†]	0.47	0.84	0.83	0.58	0.68

[†] Approaches with failure recovery capabilities.

Table 1: Benchmark results on MAP-THOR and RECAP-THOR using a three-robot setup.



Thank You!